SSOAP − A USEPA Toolbox for Sanitary Sewer Overflow Analysis and Control Planning

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ABSTRACT

The United States Environmental Protection Agency (USEPA) has identified a need to use proven methodologies to develop computer tools that help communities properly characterize rainfall-derived infiltration and inflow (RDII) into sanitary sewer systems and develop sanitary sewer overflow (SSO) control plans.

To accomplish this goal, the USEPA entered into a cooperative research and development agreement (CRADA) in 2002 with a technology partner − Camp Dresser & McKee (CDM) Inc. − to develop a public-domain Sanitary Sewer Overflow Analysis and Planning (SSOAP) Toolbox. The CRADA also prepared a technical guide for analyzing RDII, performing capacity analyses of sanitary sewer systems, and developing SSO control plans using the toolbox. This paper describes the CRADA outcomes and individual tools in the SSOAP Toolbox to perform capacity analyses of sanitary sewer systems and develop SSO control plans.

INTRODUCTION

A properly designed, operated and maintained sanitary sewer system is meant to collect and convey all of the sewage that flows into it to a wastewater treatment plant. However, occasional unintentional discharges of raw sewage from municipal sanitary sewers to streets, private property, basements and receiving waters – called sanitary sewer overflows (SSOs) – occur in many systems.

RDII into sanitary sewer systems has long been recognized as a major source of operating problems in sanitary sewer systems across the United States. Although sanitary sewer systems are generally designed to accommodate rainfall-derived infiltration and inflow (RDII) during wet weather, these flows often exceed the design allowances. When this occurs, operational problems such as basement flooding, manhole overflows, bypasses to storm sewers, and direct discharges to receiving waters from SSOs often result. The USEPA regards these problems,

especially SSOs, as a high priority for corrective action by system owners and operators.

Available data indicates that essentially all collection systems experience periodic SSOs. Additionally, between one-third and two-thirds of the nation's sanitary sewer systems have problems with SSOs or excessive peak flows at the wastewater treatment plant (WWTP). The national cost to mitigate SSOs for the next 20-year period was estimated at $164 billion (2007 dollars) to attain zero overflows per year per municipality and about $119 billion and $96 billion, respectively, to attain one and two overflows per year per municipality (USEPA 1996).

Regulatory Framework

On August 26, 2004, the USEPA delivered to Congress a report (USEPA 2004) on the impacts and control of combined sewer overflows (CSOs) and SSOs. The report indicates that the occurrence of combined sewer overflows (CSOs) and SSOs is widespread, and between 23,000 and 75,000 SSOs occur each year in the United States, resulting in releases of between 3 billion and 10 billion gallons of untreated wastewater. These events cause or contribute to environmental and human health impacts. Further, the report indicates that there are many existing structural and non-structural technologies that are well-suited for SSO control.

Since 1999, the USEPA has focused on SSO problems with compliance assistance and enforcement activities according to the "Compliance and Enforcement Strategy Addressing Combined Sewer Overflows and Sanitary Sewer Overflows," issued April 27, 2000 (USEPA 2000). In addition, the USEPA has been evaluating options for improving NPDES permit requirements for municipal sanitary sewer systems.

Although there is no national regulatory program specific to SSOs, a number of USEPA regions and state agencies have initiated programs to address SSOs. The USEPA Region 4's Management, Operation, and Maintenance (MOM) Program is one example, along with state programs such as those in California, Oklahoma and North Carolina. As these and other programs are implemented, along with updated NPDES permit requirements placed on sanitary sewer systems, the standard methodologies and planning tools will be useful for system owners/operators in addressing SSOs.

Technical Challenge

Infiltration/inflow (I/I) problems have long been the primary focus related to sewer lines' operation and maintenance (Lai and Field 2001). Besides structural failures, the problems caused by aging and deteriorating collection systems include excessive I/I that rob capacity and negatively affect operation of the entire sewerage system. The rainfall derived extraneous flows, usually termed as RDII, can cause significant increases in peak flows in the system.

The three major components of wet-weather wastewater flow into a sanitary system − base wastewater flow (BWF), groundwater infiltration (GWI) and RDII − are illustrated in **Figure 1** and are discussed further.

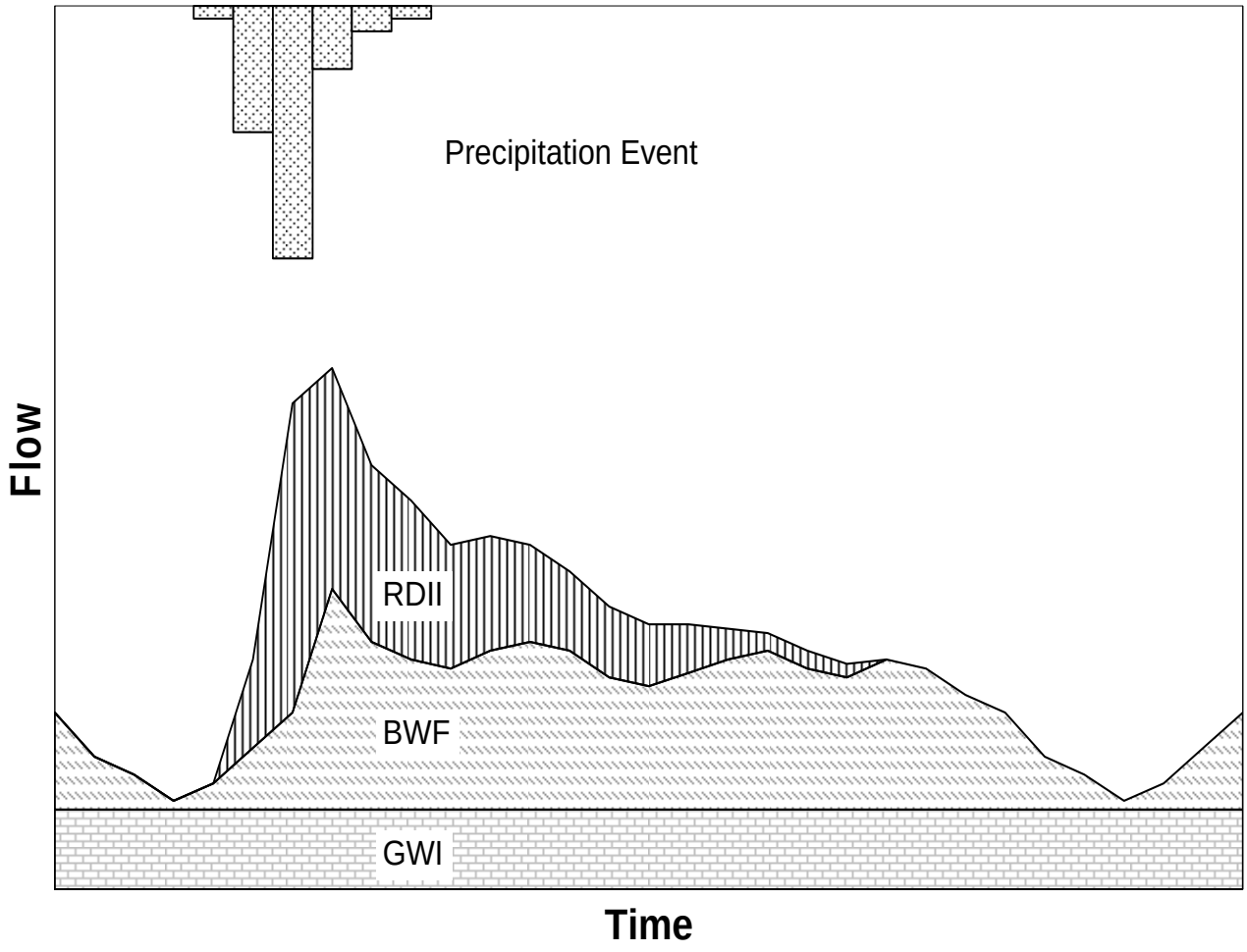


Figure 1. Three components of wet-weather wastewater flow during a precipitation event.

BWF and GWI

BWF is the residential, commercial, institutional and industrial flow that discharges to a sanitary sewer system for collection and treatment. GWI represents the infiltration of groundwater that enters a collection system through leaky pipes, pipe joints, and manhole walls. GWI varies throughout the year, often trending higher in late winter and spring as groundwater levels and soil moisture levels rise, and subsiding in late summer or after an extended dry period.

Together, GWI and BWF comprise the dry-weather flow (DWF) that occurs in a sanitary sewer system. Because the determination of GWI and BWF components of DWF is not an exact science, various assumptions related to the water consumption

return rates and wastewater monitored during early morning hours are typically used to help estimate these flows components.

RDII

In most systems, RDII is the major component of peak wastewater flows during wet weather and is typically responsible for capacity-related SSO problems and basement backups. Snowmelt may also contribute to RDII flows. **Figure 2** depicts various pathways of RDII into sanitary sewer systems. "Inflow" is the water that enters a sanitary sewer system directly via depressed manhole lids and frames, downspouts, sump pumps, foundation drains, areaway drains, and cross-connections with storm sewers. Although direct connections such as downspouts, sump pumps, foundation drains, and areaway drains are no longer common design practices, they still exist and contribute to inflow in many older systems. Inflow typically occurs shortly after a rainfall starts and stops quickly once it stops. Inflow is typically the major component of the RDII peak flow.

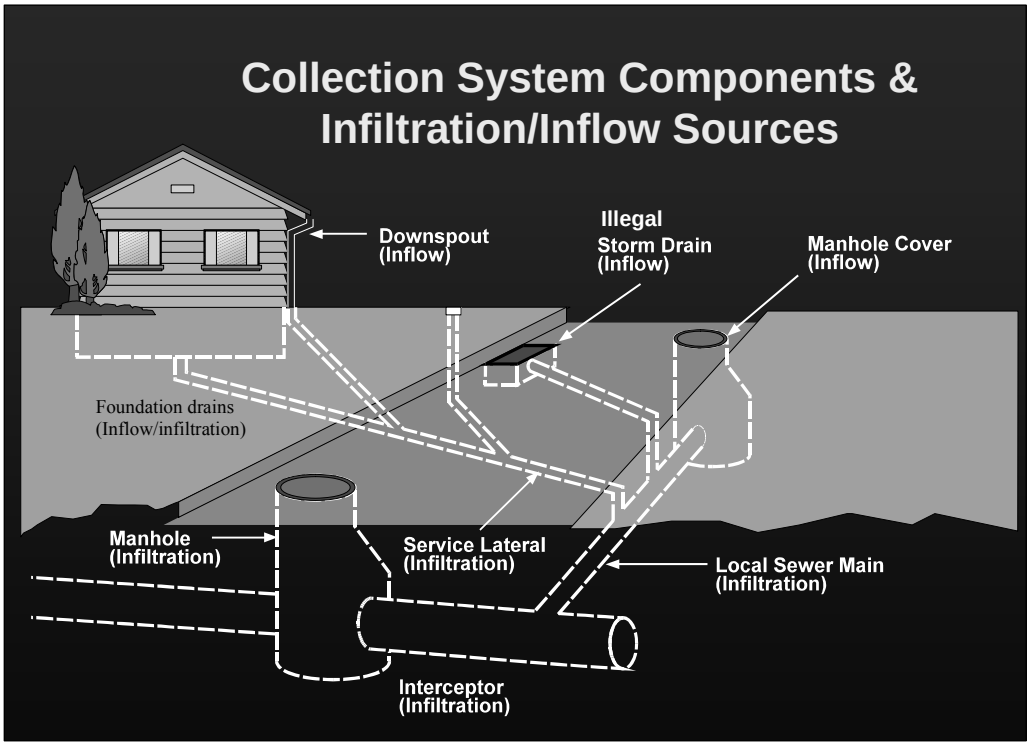


Figure 2. Pathways of I/I into sanitary sewer systems.

Rainfall-derived infiltration (RDI) refers to rainfall runoff that filters through the soil before entering a sanitary sewer system through damaged pipe sections, leaky joints or poor manhole connections. These defects can occur in both the public right-of-way portions of the sanitary sewer system or in individual service laterals on private property. Infiltration processes typically extend beyond the end of rainfall and take some time to recede to zero after the storm event. A system may experience a fast RDI response, a slow RDI response or both.

The response of a sanitary sewer system is quite complex. Various factors control RDII responses in addition to the rainfall (volume, intensity and duration) and antecedent moisture conditions, including depth to groundwater, depth to bedrock, land slope, number and size of sewer system defects, type of storm drainage system, soil characteristics, and type of sewer backfill. Further, RDII responses can vary greatly due to spatial rainfall distributions over a sewershed.

SSO CRADA AND OUTCOMES

The USEPA has identified a need to use proven methodologies to develop computer tools to assist communities in developing reliable RDII estimates and SSO control plans that are in line with their projected annual capital budgets and provide flexibility in future improvements. As part of a CRADA, USEPA and CDM developed public-domain software tools and technical guidance to support SSO control planning. The following are the outcomes of this CRADA:

- RDII Prediction Methodologies
- Technical Report
- Software tools − the SSOAP Toolbox

The USEPA reviewed prevailing industry practices for RDII prediction and selected methods that are proven for a range of application needs, The Technical Report describes the RDII prediction methodologies and the SSOAP Toolbox developed based on these methodologies.

Overview of RDII Prediction Methodologies

The complex nature of RDII sources and its pathway to the sanitary system poses a significant technical challenge to accurately and reliably predict RDII over a range of rainfall and soil moisture conditions.

A Water Environment Research Foundation (WERF) report (Bennett et al. 1999) listed and evaluated eight broad categories of RDII prediction methods. The WERF study concluded that all RDII prediction methods should be used with the site-specific database of rainfall and flow observations during both wet and dry periods. However, no one method was likely to be universally applicable because of a variety of site conditions and analysis application needs. The report concluded that the synthetic unit hydrograph (SUH) and rainfall/flow regression were preferred for predicting peak flows and event volumes for single as well as multiple-storm events. The ability of a good multiple-storm peak and volume prediction is important in extrapolating data beyond the calibration events for a prolonged period simulation, which is essential for evaluating the effect of RDII on storage and treatment requirements.

The RTK method depicted in **Figure 3**, as used in Storm Water Management Model Version 4 (Huber and Dickenson 1988) and Version 5 (SWMM5) (USEPA 2007), is probably the most popular SUH method. This method uses three triangular unit hydrographs to represent the various ways that precipitation contributes to RDII. Also

as shown on Figure 3, the shape of each of three unit hydrographs is defined by three parameters: R, T and K. R represents the fraction of rainfall falling on the sewered area that enters the sewer system as RDII. T is the time to the peak RDII flow in hours, and K is the ratio of the time of recession to the time of peak. The RDII volumes of three unit hydrographs are designated as R_1, R_2, and R_3. The sum of R_1, R_2, and R_3 equals the total R-value for the storm event that represents the total fraction of rainfall over the sewershed that entered the sewer system. A high R_1 value indicates that the RDII is primarily inflow driven. If more of the total R-value is allocated to R_2 and R_3, this indicates that the RDII is primarily infiltration driven. This knowledge is useful during a sewer system evaluation survey to determine whether a point repair or a comprehensive rehabilitation approach is more suitable.

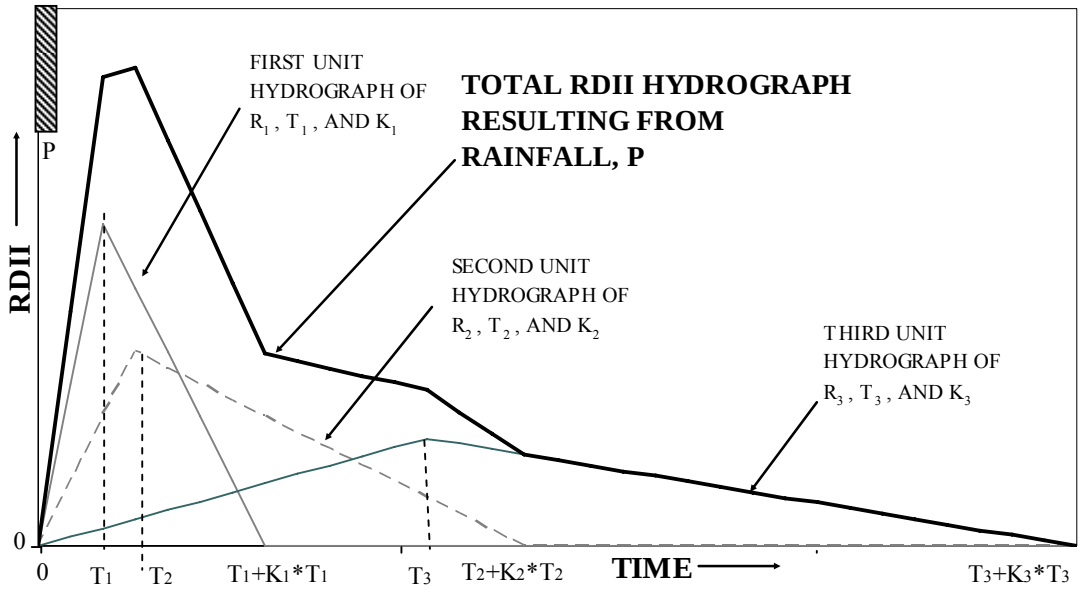


Figure 3. The RTK method parameters and summation of three unit hydrographs.

The RTK method was developed and enhanced by CDM staff members (Giguere and Riek 1983; CDM *et al*. 1985; Miles *et al*. 1996; Vallabhaneni *et al*., 2002). It has been used on sewer system master planning projects throughout the country by CDM engineers and their municipal agency staff members since the mid-1980s. This CRADA is intended for CDM to share its rich and long experience in applying the RTK method for SSO analysis and control planning. Hence, the SSOAP Toolbox initially incorporated the RTK method only. Other RDII prediction methods may be included in future effort to expand the SSOAP Toolbox by USEPA or the private sector.

Overview of the Technical Report

The USEPA published the technical report as a guide to provide the Toolbox users with technical information needed for its effective use for analysis and mitigation of SSO-related problems (USEPA 2007). The technical report is not intended to serve as

the user manual for the SSOAP Toolbox, which is a separate document included with the software package. Instead, the report provides an introductory hydrologic approach and identifies a RDII methodology for initial incorporation into the SSOAP Toolbox; an overview of the required sewer system hydraulic analysis; and data collection requirements to support SSO planning and analysis using the SSOAP Toolbox. In addition, the report describes the tools and their functions for performing a sanitary sewer system capacity assessment.

The report also includes a description of the application of the USEPA's SWMM5 application within the SSOAP Toolbox for assessing the baseline hydraulic conditions of the system and quantifying capacity improvements of various identified improvement scenarios. Guidance is provided for establishing system improvement objectives, screening potential options for improvements, developing improvement scenarios and using SWMM5 model output to evaluate alternatives. Finally, the report provides a case study that demonstrates how the RDII methodology described in this technical report has been effectively used in SSO planning and analysis.

Overview of the SSOAP Toolbox

Figure 4 shows the overall structure of the SSOAP Toolbox and the flow of data through the analysis and planning process. The Toolbox integrates an existing database of a sanitary sewer system and contains five functional tools for: Database Management, RDII Analysis, RDII Hydrograph Generation, SSOAP-SWMM5

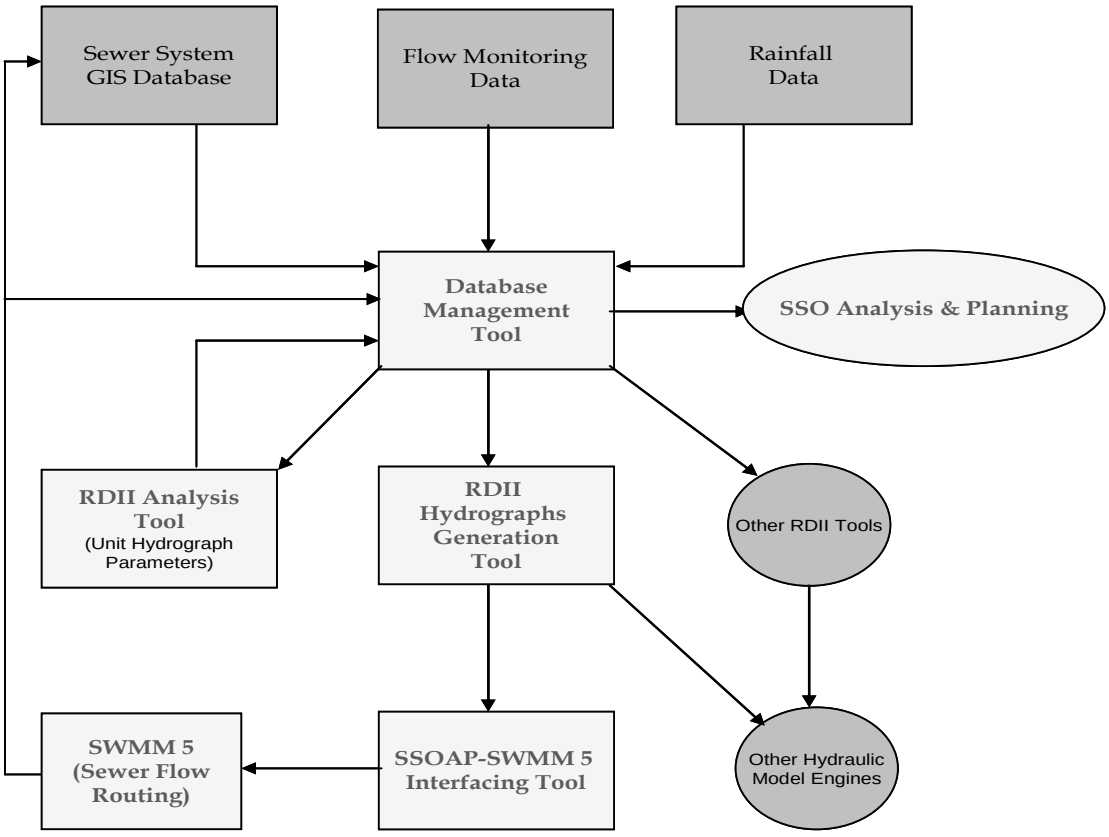


Figure 4. System flow chart of the SSOAP Toolbox.

Interface, and Sewer Flow Routing (SWMM5). The USEPA plans to release a beta version of the Toolbox along with user's manual to the public in 2008. The individual components of the SSOAP tools are described below.

Database Management Tool (DMT)

The DMT serves as the command center of the Toolbox. It stores and organizes data using a standard Microsoft Access® database called SSOAP System Database (SSD). The DMT organizes data in the SSD, interfaces with several external data sources and manages the import of data from external databases into the SSD and export of data from it. The external data sources may include sewer system GIS databases, data from flow monitoring programs, and data from rainfall monitoring programs or radar rainfall analyses, and hydraulic modeling analysis results. The other primary functions of the DMT are to provide interface with other tools in the Toolbox and manage data produced by them. To facilitate the data management function, the DMT has included three additional utilities: (1) a software to perform rainfall and flow data quality control; (2) a rainfall data analysis utility to identify wet-weather events of interest and perform rainfall analysis to determine rainfall volume, peak rainfall intensity, and antecedent moisture conditions; and (3) a scenario management utility to support sanitary sewer system capacity analysis and planning. Integrating these additional utilities into the Toolbox improves the efficiency and results of comprehensive data reviews.

RDII Analysis Tool

This tool allows users to perform the wastewater hydrograph decomposition (as depicted in Figure 3) and determine up to three sets of RDII parameters (R, T, K) in the RTK method incorporated into SSOAP as previously described. Decomposition of the flow data into each of the major wastewater components is essential for identifying the sources of flow in the system and for developing an understanding of the system RDII characteristics.

The RDII Analysis Tool has the ability to support four major analyses:

1. **DWF analysis:** Determines the representative weekday and weekend DWF hydrographs at each metered station, which will be used to perform wet-weather flow analysis.
2. **Wet-weather flow (WWF) analysis:** Determines the RDII hydrograph for a storm event by subtracting the DWF hydrograph and GWI adjustment flow from the total monitored hydrograph (BWF plus GWI plus RDII).
3. **RTK unit hydrograph curve fitting analysis**: Determines R, T, K values through a systematic analysis of measured sewer flow and rainfall for each metering location. The goal of the unit hydrograph curve fitting process is to adjust the R, T, K parameters for each event analyzed. When these values are applied to the monitored precipitation data, the simulated RDII reasonably matches the monitored flow.
4. **RTK unit hydrograph parameter predictive analysis for unmetered**

and design precipitation conditions: Allows users to perform predictive analysis of the R, T, K unit hydrograph parameters to extrapolate from that for the measured conditions to non-measured or design conditions. The tool will have the ability to perform a statistical analysis of the R, T, K parameters stored in the SSD to develop correlations between the observed precipitation conditions and the system RDII response, particularly the R-value.

RDII Hydrograph Generation Tool

This tool generates the RDII hydrograph of a sewershed for the selected rainfall events using its physical characteristics (e.g., sewer areas and land uses) data stored in the SSD and the R,T, and K values determined with the RDII Analysis Tool. Within this tool, users can view the RDII hydrograph generated before exporting the data as input to a sewer routing model. The tool has the ability to export RDII hydrographs to other hydraulic routing engines in addition to SWMM5.

SSOAP-SWMM5 Interface Tool

This tool is designed to assist users in organizing and incorporating the hydrographs generated by the RDII Hydrograph Generation Tool into the SWMM 5 input files. It will then initiate a SWMM5 simulation. After the SWMM5 run, the tool will deliver the SWMM5 simulation results to the DMT, where the model results will be organized in the SSD for additional post-processing.

SWMM5

While other sewer routing models may be linked to the DMT and used, the SSOAP Toolbox uses SWMM5 to perform the actual dynamic flow routing through a sewer network system. It also uses the graphic utility interface capability in SWMM5 to visualize the sewer system responses and selectively exports the output data for further analysis.

SUMMARY AND CONCLUSIONS

Because of the serious and growing concern over the health and environmental risks of SSOs, the USEPA entered into a CRADA in 2002 with CDM as a technology partner to develop a public domain computer analysis and modeling toolbox to assist SSO communities in quantifying RDII in their system and developing SSO mitigation plans. The Toolbox is named SSO Analysis and Planning (SSOAP). It contains a suite of computer software tools to facilitate the analysis for quantification of RDII and performance of capacity analysis in sanitary sewer systems. It is designed to provide technical support for complying with the “C” (for Capacity) in the Capacity, Management, Operation and Maintenance (CMOM) program. The CRADA also includes a recently published USEPA technical report (USEPA 2007) and a SSOAP user’s manual (currently being developed) to guide the application of the Toolbox. The Toolbox is intended for users with appropriate education and training in hydrologic and hydraulic principles and background on sanitary sewer systems. The USEPA plans to release a beta version of the Toolbox, along with the user’s manual, to the public in 2008.

DISCLAIMER

Any opinions expressed in this paper are those of the author(s) and do not necessarily, reflect the official positions and policies of the USEPA. Any mention of products or trade names does not constitute recommendation for use by the USEPA.

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